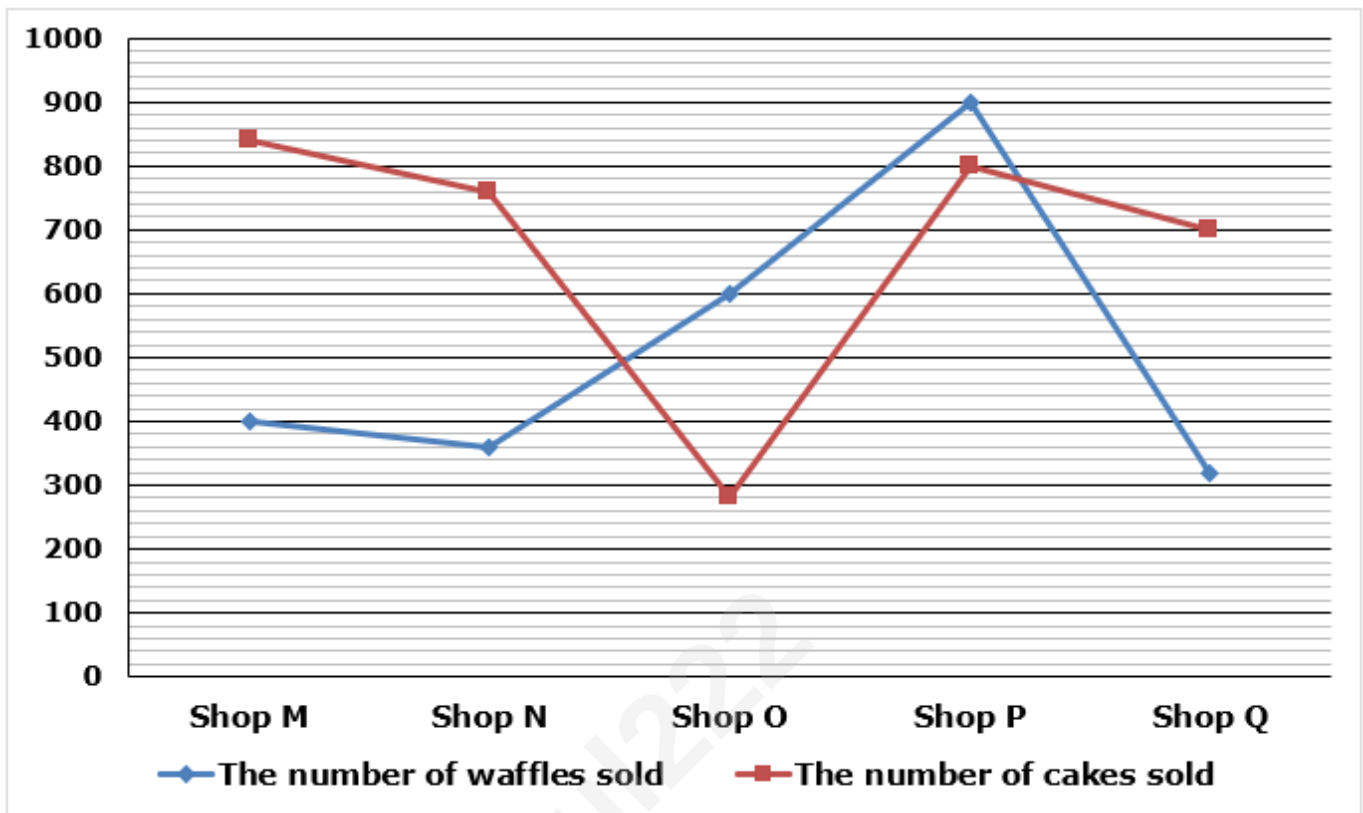


1. Questions

Study the following information carefully and answer the questions.

The given line graph shows the number of waffles sold in five different shops namely M, N, O, P and Q and also the number of cakes sold in these five shops.



Out of the total number of waffles sold in Shop P, 25% were chocolate waffles. If the number of vanilla waffles sold in Shop O is 40% less than that of Shop P, then find the number of chocolate waffles sold in Shop O.

(Only two types of waffles are sold: chocolate and vanilla)

- a. 195
- b. 205
- c. 225
- d. 180
- e. 170

2. Questions

The number of waffles sold in Shop R is 25% more than that of Shop Q and the number of cakes sold in Shop R is 380 less than that of Shop N. Find the sum of the number of waffles and cakes sold in Shop R.

- a. 700
- b. 720

- c. 740
- d. 780
- e. 760

3. Questions

The sum of the number of cookies sold in Shops O, P and Q is 220 less than the total number of waffles and cakes sold in Shop N and the ratio of waffles to cookies sold in Shop O is 2:1. Find the sum of the number of cookies sold in Shops P and Q.

- a. 800
- b. 600
- c. 400
- d. 500
- e. 700

4. Questions

Find the difference between the number of cakes sold in Shop M and Shop P.

- a. 10
- b. 50
- c. 40
- d. 30
- e. 20

5. Questions

The number of cakes sold in Shop O is what percentage of the number of waffles sold in Shop M?

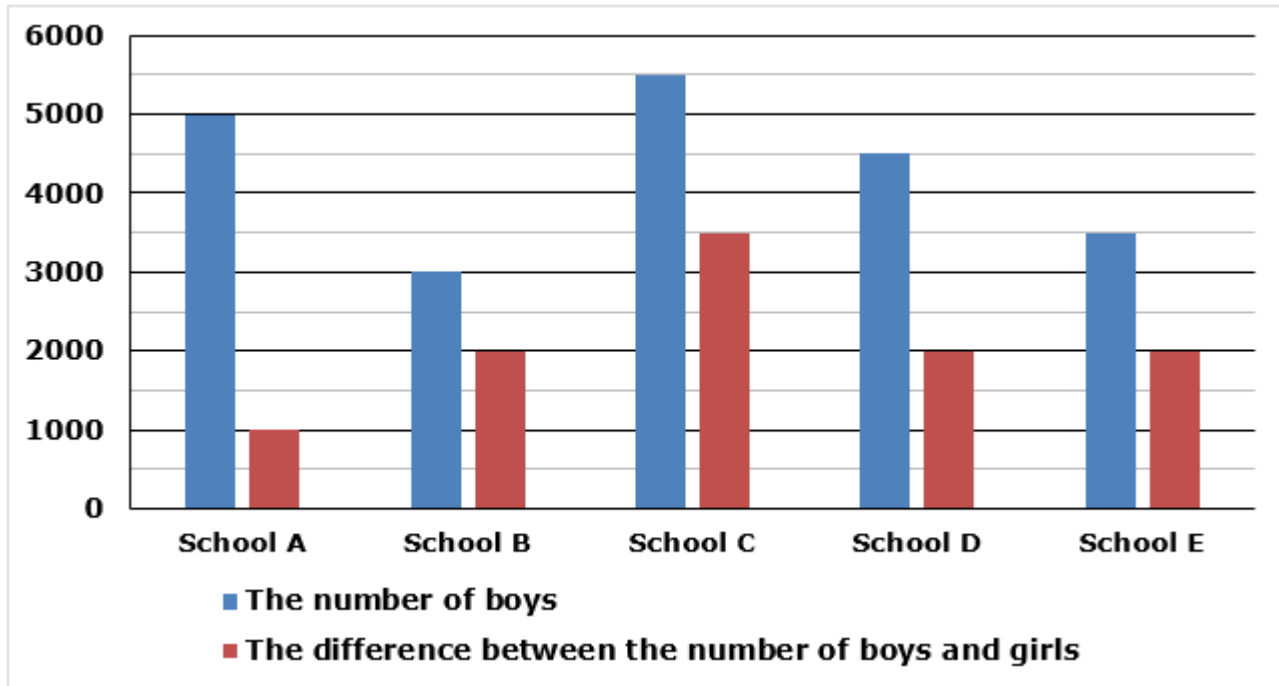
- a. 60%
- b. 80%
- c. 40%
- d. 50%
- e. 70%

6. Questions

Read the following information carefully and answer the questions.

The given bar graph shows the number of boys in five different schools namely A, B, C, D and E and also given the difference between the number of boys and girls in these five schools.

Note: The number of boys in each school is more than the number of girls.



Find the difference between the total number of students in School C and the number of girls in School D.

- a. 2000
- b. 7000
- c. 5000
- d. 4500
- e. 3000

7. Questions

Find the average of the number of girls in all the five schools together.

- a. 2200
- b. 2000
- c. 2100
- d. 2300
- e. 2400

8. Questions

If the total number of students in School F is 25% more than that of School D and the number of boys in School A is 25% more than that of School F, then find the number of girls in School F.

- a. 4000
- b. 4250

- c. 4350
- d. 4485
- e. 4750

9. Questions

If 32 students and 50 students are studying in each class of School B and C respectively, then find the ratio of the total number of classes in School C to School B.

- a. 6:5
- b. 2:5
- c. 5:6
- d. 3:2
- e. 4:3

10. Questions

The ratio of the number of boys to girls from City X in School E is 3:1 and the number of girls from City Y is $\frac{1}{2}$ of the number of boys from City Y in School E. Find the number of students from City Y in School E. (The students are only from City X and City Y)

- a. 2000
- b. 1500
- c. 3500
- d. 3000
- e. 2500

11. Questions

A vessel contains a mixture of milk and water in the ratio of 2:3 and 20% of the mixture is removed from the vessel and x litres of milk is added, then the ratio of the milk to water becomes 5:4. Find the value of x, if the initial quantity of water was 32 litres more than the quantity of milk added.

- a. 28
- b. 20
- c. 15
- d. 25
- e. 30

12. Questions

Raju bought 10 pens and 10 pencils for a total cost of Rs.300. He sold 5 pens and 6 pencils for a total cost of Rs.125 and Rs.90 respectively. If the cost price of each pen is 100% more than that of each pencil, then find the profit percentage obtained by selling one pen.

- a. 15%
- b. 18%
- c. 30%
- d. 20%
- e. 25%

13. Questions

A and B started a business together by investing Rs.x and Rs.(x + 1000) respectively and invested for 18 months and 12 months respectively. The profit obtained by A is 3 times his investment and the profit obtained by B is 150% more than the investment of A. Find the value of x.

- a. 2000
- b. 6000
- c. 4000
- d. 5000
- e. 3500

14. Questions

The ratio of the number of students in school A to B is 4:5 and the ratio of the number of students in schools A and C together to the number of students in school B is 2:1. If the total number of students in schools A, B and C is 15000, then find the number of students in school C.

- a. 4000
- b. 5000
- c. 7000
- d. 3000
- e. 6000

15. Questions

A and B invested Rs.10000 and Rs.15000 in SI at the rate of interest of x% for x/2 years and the sum of the interest obtained by A and B is Rs.50000. If A invested Rs.5000 at the rate of interest of 10% for x years, then find the interest obtained by A.

- a. Rs.20000
- b. Rs.25000

- c. Rs.30000
- d. Rs.15000
- e. Rs.10000

16. Questions

The sum of the distance covered by the boat in 14 hours in downstream and 12 hours in upstream is 530 km and the time taken by the boat to cover 260 km in still water is 13 hours. Find the downstream speed of the boat.

- a. 30 km/hr
- b. 25 km/hr
- c. 35 km/hr
- d. 15 km/hr
- e. 40 km/hr

17. Questions

A bag contains 3 blue balls, x white balls, 5 yellow balls and 4 green balls. The average number of balls in the bag is 4. If three balls are selected at random, then find the probability of selecting at least one white ball.

- a. $19/28$
- b. $23/28$
- c. $15/28$
- d. $17/28$
- e. $11/28$

18. Questions

The sum of the present ages of A and B is equal to the sum of the ages of B and C after 2 years. Four years ago, the ratio of the ages of A, B and C was 4:6:3 respectively. Find the present age of B.

- a. 20 years
- b. 12 years
- c. 24 years
- d. 28 years
- e. 16 years

19. Questions

The ratio of the length to breadth of the rectangle is 10:9 and the perimeter of the rectangle is 76

m. If the area of the rectangle is 10 m^2 more than the area of the right angled triangle and the breadth of the right angled is 28 m, then find the height of the right angled triangle.

- a. 25 m
- b. 20 m
- c. 12 m
- d. 30 m
- e. 28 m

20. Questions

The time taken by A and B together to complete a work is $(20/3)$ days and the time taken by B and C together to complete the work is 12 days and the ratio of the efficiency of A to C is 3:1. Find the time taken by B alone to complete the whole work.

- a. 18 days
- b. 20 days
- c. 10 days
- d. 30 days
- e. 24 days

21. Questions

What value should come in the place of (?) in the following questions?

$$8/7 \text{ of } 343 + 288 - ? * 2 = 12^2$$

- a. 268
- b. 264
- c. 272
- d. 252
- e. 250

22. Questions

$$\sqrt{3600} * 3 \div 4 = \sqrt{?}$$

- a. 5625
- b. 1225
- c. 4225
- d. 3025

e. 2025

23. Questions

$$982 * 8 - 432 * 4 - 612 * 5 = ?$$

- a. 2222
- b. 3004
- c. 3068
- d. 3245
- e. 2842

24. Questions

$$8(5/11) - 4(9/11) = ? \div (2(3/4))$$

- a. 10
- b. 15
- c. 18
- d. 21
- e. 24

25. Questions

$$(? + 50 \div 10) * 40 = 160$$

- a. -2
- b. 2
- c. -3
- d. -1
- e. 1

26. Questions

What approximate value should come in the place of (?) in the following questions?

$$(\sqrt{255} * \sqrt{291}) \div \sqrt{402} = \sqrt{15.8} * ? \div 24.95$$

- a. 75
- b. 70
- c. 65
- d. 85

e. 80

27. Questions

$$?^3 + 363.43 = 79.85\% \text{ of } 3199.72$$

- a. 15
- b. 11
- c. 13
- d. 19
- e. 17

28. Questions

$$3780.43 \div 11.85 + 14.89^2 = ? * 4.95$$

- a. 104
- b. 116
- c. 124
- d. 108
- e. 112

29. Questions

$$\sqrt{?} * 39.99\% + 13.73^2 = \sqrt{39999}$$

- a. 225
- b. 64
- c. 144
- d. 289
- e. 100

30. Questions

$$19.9\% \text{ of } 110.31 + 11.96\% \text{ of } 149.76 - 30.18 = ?$$

- a. 10
- b. 15
- c. 20
- d. 18
- e. 13

31. Questions

What value should come in the place of (?) in the following number series?

40, 126, 257, 443, 694, ?

- a. 1080
- b. 1060
- c. 1040
- d. 1020
- e. 1120

32. Questions

8, 20, 36, ?, 80, 108

- a. 44
- b. 58
- c. 56
- d. 50
- e. 48

33. Questions

44, 71, 196, 539, ?, 4067

- a. 1878
- b. 1873
- c. 1892
- d. 1847
- e. 1870

34. Questions

500, 250, 250, ?, 750, 1875

- a. 385
- b. 380
- c. 375
- d. 360
- e. 345

35. Questions

1, 89, ?, 78, 23, 67

- a. 12
- b. 10
- c. 15
- d. 18
- e. 20

36. Questions

Find out the wrong number in the following number series.

4, 6, 15, 48, 201, 1011

- a. 15
- b. 1011
- c. 48
- d. 4
- e. 6

37. Questions

2280, 2230, 2190, 2130, 2060, 1980

- a. 2060
- b. 2230
- c. 2280
- d. 2190
- e. 2130

38. Questions

216, 64, 100, 144, 2744, 256

- a. 216
- b. 256
- c. 144
- d. 64
- e. 100

39. Questions**54, 93, 171, 288, 440, 639**

- a. 440
- b. 171
- c. 639
- d. 54
- e. 93

40. Questions**68, 84, 100, 164, 420, 1444**

- a. 164
- b. 1444
- c. 100
- d. 68
- e. 84

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give the answer as,

I). $x^2 - 14x + 45 = 0$

II). $y^2 - 26y + 88 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship can't be determined
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $4x^2 - 16x + 15 = 0$

II). $y^2 - 7y + 12 = 0$

- a. $x > y$
- b. $x \geq y$

- c. $x = y$ or relationship can't be determined
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $x^2 - 30x + 224 = 0$

II). $y^2 - 22y + 120 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship can't be determined
- d. $x < y$
- e. $x \leq y$

44. Questions

I). $x^2 - 34x + 288 = 0$

II). $y^2 - 25y + 144 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship can't be determined
- d. $x < y$
- e. $x \leq y$

45. Questions

I). $x^2 + 15x + 54 = 0$

II). $y^2 - 7y - 78 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship can't be determined
- d. $x < y$
- e. $x \leq y$

Explanations:

1. Questions

Answer: A

The number of vanilla waffles sold in Shop P = $900 * 75/100 = 675$

The number of vanilla waffles sold in Shop O = $675 * 60/100 = 405$

The number of chocolate waffles sold in Shop O = $600 - 405 = 195$

2. Questions

Answer: D

The number of waffles sold in Shop R = $320 * 125/100 = 400$

The number of cakes sold in Shop R = $760 - 380 = 380$

Required sum = $400 + 380 = 780$

3. Questions

Answer: B

The sum of the number of cookies sold in Shops O, P and Q = $360 + 760 - 220 = 1120 - 220 = 900$

The number of cookies sold in Shop O = $600 * 1/2 = 300$

The sum of the number of cookies sold in Shops P and Q = $900 - 300 = 600$

4. Questions

Answer: C

Required difference = $840 - 800 = 40$

5. Questions

Answer: E

Required percentage = $280/400 * 100 = 70\%$

6. Questions

School A:

The number of boys = 5000

The number of girls = $5000 - 1000 = 4000$

The total number of students = $5000 + 4000 = 9000$

School B:

The number of boys = 3000

The number of girls = $3000 - 2000 = 1000$

The total number of students = $3000 + 1000 = 4000$

School C:

The number of boys = 5500

The number of girls = $5500 - 3500 = 2000$

The total number of students = $5500 + 2000 = 7500$

School D:

The number of boys = 4500

The number of girls = $4500 - 2000 = 2500$

The total number of students = $4500 + 2500 = 7000$

School E:

The number of boys = 3500

The number of girls = $3500 - 2000 = 1500$

The total number of students = $3500 + 1500 = 5000$

Answer: C

Required difference = $7500 - 2500 = 5000$

7. Questions**School A:**

The number of boys = 5000

The number of girls = $5000 - 1000 = 4000$

The total number of students = $5000 + 4000 = 9000$

School B:

The number of boys = 3000

The number of girls = $3000 - 2000 = 1000$

The total number of students = $3000 + 1000 = 4000$

School C:

The number of boys = 5500

The number of girls = $5500 - 3500 = 2000$

The total number of students = $5500 + 2000 = 7500$

School D:

The number of boys = 4500

The number of girls = $4500 - 2000 = 2500$

The total number of students = $4500 + 2500 = 7000$

School E:

The number of boys = 3500

The number of girls = $3500 - 2000 = 1500$

The total number of students = $3500 + 1500 = 5000$

Answer: A

Required average = $(4000 + 1000 + 2000 + 2500 + 1500)/5 = 11000/5 = 2200$

8. Questions**School A:**

The number of boys = 5000

The number of girls = $5000 - 1000 = 4000$

The total number of students = $5000 + 4000 = 9000$

School B:

The number of boys = 3000

The number of girls = $3000 - 2000 = 1000$

The total number of students = $3000 + 1000 = 4000$

School C:

The number of boys = 5500

The number of girls = $5500 - 3500 = 2000$

The total number of students = $5500 + 2000 = 7500$

School D:

The number of boys = 4500

The number of girls = $4500 - 2000 = 2500$

The total number of students = $4500 + 2500 = 7000$

School E:

The number of boys = 3500

The number of girls = $3500 - 2000 = 1500$

The total number of students = $3500 + 1500 = 5000$

Answer: E

The total number of students in School F = $7000 * 125/100 = 8750$

The number of boys in School F = $5000 * 100/125 = 4000$

The number of girls in School F = $8750 - 4000 = 4750$

9. Questions

School A:

The number of boys = 5000

The number of girls = $5000 - 1000 = 4000$

The total number of students = $5000 + 4000 = 9000$

School B:

The number of boys = 3000

The number of girls = $3000 - 2000 = 1000$

The total number of students = $3000 + 1000 = 4000$

School C:

The number of boys = 5500

The number of girls = $5500 - 3500 = 2000$

The total number of students = $5500 + 2000 = 7500$

School D:

The number of boys = 4500

The number of girls = $4500 - 2000 = 2500$

The total number of students = $4500 + 2500 = 7000$

School E:

The number of boys = 3500

The number of girls = $3500 - 2000 = 1500$

The total number of students = $3500 + 1500 = 5000$

Answer: A

The total number of classes in School B = $4000/32 = 125$

The total number of classes in School C = $7500/50 = 150$

Required ratio = $150:125 = 6:5$

10. Questions

School A:

The number of boys = 5000

The number of girls = $5000 - 1000 = 4000$

The total number of students = $5000 + 4000 = 9000$

School B:

The number of boys = 3000

The number of girls = $3000 - 2000 = 1000$

The total number of students = $3000 + 1000 = 4000$

School C:

The number of boys = 5500

The number of girls = $5500 - 3500 = 2000$

The total number of students = $5500 + 2000 = 7500$

School D:

The number of boys = 4500

The number of girls = $4500 - 2000 = 2500$

The total number of students = $4500 + 2500 = 7000$

School E:

The number of boys = 3500

The number of girls = $3500 - 2000 = 1500$

The total number of students = $3500 + 1500 = 5000$

Answer: D

The number of boys from City X in School E = $3m$

The number of girls from City X in School E = m

The number of boys from City Y in School E = $2n$

The number of girls from City Y in School E = n

$$3m + 2n = 3500 \text{ ---- (1)}$$

$$m + n = 1500 \text{ ---- (2)}$$

From equation (1) and (2),

$$m = 500$$

$$n = 1000$$

The number of boys from City Y in School E = $2 * 1000 = 2000$

The number of girls from City Y in School E = 1000

Required sum = $2000 + 1000 = 3000$

11. Questions

Answer: A

The initial quantity of milk in the vessel = $2a$

The initial quantity of water in the vessel = $3a$

The quantity of the mixture removed = $5a * 20/100 = a$

$$(2a - a * 2/5 + x)/(3a - a * 3/5) = 5/4$$

$$(10a - 2a + 5x)/(15a - 3a) = 5/4$$

$$(8a + 5x)/12a = 5/4$$

$$8a + 5x = 15a$$

$$7a = 5x$$

$$a = 5x/7$$

$$3a - x = 32$$

$$3 * 5x/7 - x = 32$$

$$15x - 7x = 32 * 7$$

$$8x = 32 * 7$$

$$x = 28$$

12. Questions

Answer: E

Let the cost price of each pencil be Rs.a.

The cost price of each pen = $a * 200/100 = Rs.2a$

$$10 * 2a + 10 * a = 300$$

$$30a = 300$$

$$a = 10$$

The cost price of each pen = $2 * 10 = Rs.20$

The selling price of each pen = $125/5 = Rs.25$

The profit percentage obtained = $(25 - 20)/20 * 100 = 5/20 * 100 = 25\%$

13. Questions

Answer: C

The profit obtained by A = $3x$

The profit obtained by B = $x * 250/100 = 2.5x$

$$x * 18/((x + 1000) * 12) = 3x/2.5x$$

$$x * 3/((x + 1000) * 2) = 3/2.5$$

$$x/(2x + 2000) = 1/2.5$$

$$2.5x = 2x + 2000$$

$$0.5x = 2000$$

$$x = 4000$$

14. Questions

Answer: E

The number of students in School A = $4x$

The number of students in School B = $5x$

The number of students in schools A and C together = $2 * 5x = 10x$

The number of students in School C = $10x - 4x = 6x$

$$4x + 5x + 6x = 15000$$

$$15x = 15000$$

$$x = 1000$$

The number of students in School C = $6 * 1000 = 6000$

15. Questions

Answer: E

$$SI = PNR/100$$

$$10000 * x/100 * x/2 + 15000 * x/100 * x/2 = 50000$$

$$50x^2 + 75x^2 = 50000$$

$$125x^2 = 50000$$

$$x^2 = 400$$

$$x = 20$$

The interest obtained by A = $5000 * 20 * 10/100 = \text{Rs.}10000$

16. Questions

Answer: B

The speed of the boat in still water = $260/13 = 20 \text{ km/hr}$

Let the speed of the stream be $C \text{ km/hr}$.

$$(20 + C) * 14 + (20 - C) * 12 = 530$$

$$280 + 14C + 240 - 12C = 530$$

$$520 + 2C = 530$$

$$2C = 10$$

$$C = 5 \text{ km/hr}$$

The downstream speed of the boat = $20 + 5 = 25$ km/hr

17. Questions

Answer: D

The total number of balls in the bag = $4 * 4 = 16$

The sum of the number of blue, yellow and green balls = $3 + 5 + 4 = 12$

The number of white balls in the bag = $16 - 12 = 4$

Required probability = $(4C_1 * 12C_2 + 4C_2 * 12C_1 + 4C_3)/16C_3$

= $(4 * 12 * 11/2 + 4 * 3 * 12/2 + 4 * 3 * 2 / (3 * 2)) / ((16 * 15 * 14) / (3 * 2 * 1))$

= $(264 + 72 + 4) / 560 = 340 / 560 = 17/28$

18. Questions

Answer: D

The present age of A = $(4x + 4)$ years

The present age of B = $(6x + 4)$ years

The present age of C = $(3x + 4)$ years

$4x + 4 + 6x + 4 = (6x + 4 + 2) + (3x + 4 + 2)$

$10x + 8 = 9x + 12$

$x = 4$

The present age of B = $6 * 4 + 4 = 24 + 4 = 28$ years

19. Questions

Answer: A

The length of the rectangle, $l = 10x$

The breadth of the rectangle, $b = 9x$

The perimeter of the rectangle = $2(l + b)$

$2 * (10x + 9x) = 76$

$38x = 76$

$x = 2$ m

The length of the rectangle = $10 * 2 = 20$ m

The breadth of the rectangle = $9 * 2 = 18$ m

The area of the rectangle = $l * b = 20 * 18 = 360$ m²

The area of the right angled triangle = $360 - 10 = 350$ m²

The breadth of the right angled triangle, $B = 28$ m

Let the height of the right angled triangle be h.

$$\frac{1}{2} * B * h = 350$$

$$\frac{1}{2} * 28 * h = 350$$

$$h = 25 \text{ m}$$

20. Questions

Answer: B

The ratio of the time taken by A to C to complete the work = $(\frac{1}{3}) : (\frac{1}{1}) = 1:3$

The time taken by A alone to complete the work = x

The time taken by C alone to complete the work = 3x

$$\frac{1}{x} + \frac{1}{3x} = \frac{3}{20}$$

$$\frac{1}{3x} + \frac{1}{3x} = \frac{1}{12}$$

$$\frac{3}{20} - \frac{1}{x} = \frac{1}{12} - \frac{1}{3x}$$

$$\frac{3}{20} - \frac{1}{12} = \frac{1}{x} - \frac{1}{3x}$$

$$(9 - 5)/60 = (3 - 1)/3x$$

$$4/60 = 2/3x$$

$$x = 10$$

The time taken by B alone to complete the work = $\frac{3}{20} - \frac{1}{10} = (3 - 2)/20 = \frac{1}{20} = 20 \text{ days}$

21. Questions

Answer: A

$$\frac{8}{7} \text{ of } 343 + 288 - ? * 2 = 12^2$$

$$392 + 288 - ? * 2 = 144$$

$$? * 2 = 536$$

$$? = 268$$

22. Questions

Answer: E

$$\sqrt{3600} * 3 \div 4 = \sqrt{?}$$

$$60 * \frac{3}{4} = \sqrt{?}$$

$$45 = \sqrt{?}$$

$$? = 2025$$

23. Questions

Answer: C

$$982 * 8 - 432 * 4 - 612 * 5 = ?$$

$$7856 - 1728 - 3060 = ?$$

$$? = 3068$$

24. Questions**Answer: A**

$$8 \left(\frac{5}{11} \right) - 4 \left(\frac{9}{11} \right) = ? \div \left(2 \left(\frac{3}{4} \right) \right)$$

$$93/11 - 53/11 = ? \div 11/4$$

$$40/11 = ? * 4/11$$

$$? = 10$$

25. Questions**Answer: D**

$$(? + 50 \div 10) * 40 = 160$$

$$? + 5 = 4$$

$$? = -1$$

26. Questions**Answer: D**

$$(\sqrt{255} * \sqrt{291}) \div \sqrt{402} = \sqrt{15.8} * ? \div 24.95$$

$$(\sqrt{256} * \sqrt{289}) \div \sqrt{400} = \sqrt{16} * ? \div 25$$

$$16 * 17/20 = 4 * ?/25$$

$$? = 85$$

27. Questions**Answer: C**

$$?^3 + 363.43 = 79.85\% \text{ of } 3199.72$$

$$?^3 + 363 = 80/100 * 3200$$

$$?^3 + 363 = 2560$$

$$?^3 = 2197$$

$$? = 13$$

28. Questions**Answer: D**

$$3780.43 \div 11.85 + 14.89^2 = ? * 4.95$$

$$3780/12 + 15^2 = ? * 5$$

$$315 + 225 = ? * 5$$

$$540 = ? * 5$$

$$? = 108$$

29. Questions

Answer: E

$$\sqrt{?} * 39.99\% + 13.73^2 = \sqrt{39999}$$

$$40/100 * \sqrt{?} + 14^2 = \sqrt{40000}$$

$$2/5 * \sqrt{?} + 196 = 200$$

$$2/5 * \sqrt{?} = 4$$

$$\sqrt{?} = 10$$

$$? = 100$$

30. Questions

Answer: A

$$19.9\% \text{ of } 110.31 + 11.96\% \text{ of } 149.76 - 30.18 = ?$$

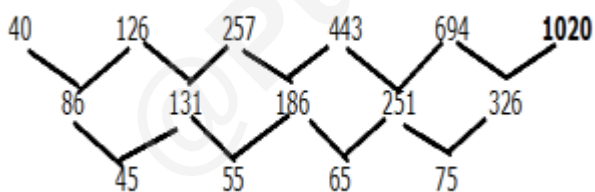
$$20/100 * 100 + 12/100 * 150 - 30 = ?$$

$$22 + 18 - 30 = ?$$

$$? = 10$$

31. Questions

Answer: D



32. Questions

Answer: C

$$8 + 12 = 20$$

$$20 + 16 = 36$$

$$36 + 20 = \mathbf{56}$$

$$56 + 24 = 80$$

$$80 + 28 = 108$$

33. Questions

Answer: E

$$44 + 3^3 = 71$$

$$71 + 5^3 = 196$$

$$196 + 7^3 = 539$$

$$539 + 11^3 = \mathbf{1870}$$

$$1870 + 13^3 = 4067$$

34. Questions

Answer: C

$$500 * 0.5 = 250$$

$$250 * 1 = 250$$

$$250 * 1.5 = \mathbf{375}$$

$$375 * 2 = 750$$

$$750 * 2.5 = 1875$$

35. Questions

Answer: A

$$1 + 88 = 89$$

$$89 - 77 = \mathbf{12}$$

$$12 + 66 = 78$$

$$78 - 55 = 23$$

$$23 + 44 = 67$$

36. Questions

Answer: C

$$4 * 1 + 2 = 6$$

$$6 * 2 + 3 = 15$$

$$15 * 3 + 4 = \mathbf{49}$$

$$49 * 4 + 5 = 201$$

$$201 * 5 + 6 = 1011$$

37. Questions

Answer: B

$$2280 - 40 = \mathbf{2240}$$

$$2240 - 50 = 2190$$

$$2190 - 60 = 2130$$

$$2130 - 70 = 2060$$

$$2060 - 80 = 1980$$

38. Questions

Answer: E

$$6^3 = 216$$

$$8^2 = 64$$

$$\mathbf{10^3 = 1000}$$

$$12^2 = 144$$

$$14^3 = 2744$$

$$16^2 = 256$$

39. Questions

Answer: A

$$54 + 39 = 93$$

$$93 + 78 = 171$$

$$171 + 117 = 288$$

$$288 + 156 = \mathbf{444}$$

$$444 + 195 = 639$$

40. Questions

Answer: D

$$\mathbf{80 + 4 = 84}$$

$$84 + 16 = 100$$

$$100 + 64 = 164$$

$$164 + 256 = 420$$

$$420 + 1024 = 1444$$

41. Questions

Answer: C

$$x^2 - 14x + 45 = 0$$

$$x^2 - 9x - 5x + 45 = 0$$

$$x(x - 9) - 5(x - 9) = 0$$

$$x = +9, +5$$

$$y^2 - 26y + 88 = 0$$

$$y^2 - 22y - 4y + 88 = 0$$

$$y(y - 22) - 4(y - 22) = 0$$

$$y = +22, +4$$

x = y or relationship can't be determined

42. Questions

Answer: D

$$4x^2 - 16x + 15 = 0$$

$$4x^2 - 6x - 10x + 15 = 0$$

$$4x(x - 6/4) - 10(x - 6/4) = 0$$

$$x = +10/4, +6/4$$

$$y^2 - 7y + 12 = 0$$

$$y^2 - 4y - 3y + 12 = 0$$

$$y(y - 4) - 3(y - 4) = 0$$

$$y = +3, +4$$

x < y

43. Questions

Answer: A

$$x^2 - 30x + 224 = 0$$

$$x^2 - 16x - 14x + 224 = 0$$

$$x(x - 16) - 14(x - 16) = 0$$

$$x = +16, +14$$

$$y^2 - 22y + 120 = 0$$

$$y^2 - 10y - 12y + 120 = 0$$

$$y(y - 10) - 12(y - 10) = 0$$

$$y = +12, +10$$

$$x > y$$

44. Questions

Answer: B

$$x^2 - 34x + 288 = 0$$

$$x^2 - 18x - 16x + 288 = 0$$

$$x(x - 18) - 16(x - 18) = 0$$

$$x = +16, +18$$

$$y^2 - 25y + 144 = 0$$

$$y^2 - 9y - 16y + 144 = 0$$

$$y(y - 9) - 16(y - 9) = 0$$

$$y = +9, +16$$

$$x \geq y$$

45. Questions

Answer: E

$$x^2 + 15x + 54 = 0$$

$$x^2 + 9x + 6x + 54 = 0$$

$$x(x + 9) + 6(x + 9) = 0$$

$$x = -9, -6$$

$$y^2 - 7y - 78 = 0$$

$$y^2 - 13y + 6y - 78 = 0$$

$$y(y - 13) + 6(y - 13) = 0$$

$$y = -6, +13$$

$$x \leq y$$